Linux Computer Literacy Report

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1 File Manager - Directories

2 Python & pip

2.1 Installing Python

First you need a compiler like VScodium. This can be downloaded and installed from its official website or through your system's package manager and the terminal. You can run the command sudo apt install python3 to install Python on your system. Make sure to verify the installation by running python3 –version in the terminal.

2.2 Python Files

.py, .ipynb and other Python-related files are commonly used in Python development. .py files are standard Python scripts, while .ipynb files are Jupyter Notebook files that allow you to create and share documents containing live code, equations, visualizations, and narrative text.

2.3 Installing pip

To install pip, you can use the following command:

```
python -m ensurepip --upgrade
```

Note: you may have a different version of python installed just replace `python` with the appropriate version, e.g., `python3`.

2.4 Installing Packages

To install packages using pip, you can use the following command:

```
pip install "package_name"
```

If a Virtual Environment is not set up this will install the package to the site-packages directory of your Python environment.

3 Venv

3.1 Creating a Virtual Environment

A virtual environment is an isolated Python environment that allows you to manage dependencies for different projects separately. You can create a virtual environment using the following command:

```
python -m venv myenv
```

3.2 Activating a Virtual Environment

To activate the virtual environment, use the following command:

```
source myenv/bin/activate  # On Unix or MacOS
myenv\Scripts\activate     # On Windows
```

4 BASH

BASH (Bourne Again SHell) is a command-line shell used for interacting with the operating system. It allows you to execute commands, run scripts, and automate tasks efficiently. the command "- -help" after any command to get more information about its usage. or the command "man" followed by the command name to access the manual. For example:

4.1 Basic BASH Commands

```
# List files and directories
ls

# Change directory
cd /path/to/directory

# Create a new directory
mkdir new_directory

# Remove a file
rm file_name

# Remove a directory and its contents
rm -r directory_name

# Display the current directory
pwd

# stop running processes in the terminal
Ctrl+C

# copy and paste changes picking up a shift
Cntrl+Shift+V  # Paste in terminal
Cntrl+Shift+C  # Copy in terminal

# stop the terminal
exit

# clear the terminal
clear

# run as administrator
sudo command_here

# search for a file or directory
find path -name file_name

% open vscodium
codium . # Open the current directory in VScodium
codium file_name  # Open a specific file in VScodium
```

4.2 Creating Bash Shortcuts /.bashrc file

I have added an example /.bashrc file in the attached .txt file. In linux you can add a /.bashrc file which you can use to create custom aliases and functions to make your terminal workflow more efficient. This makes setting up a new project very easy, for example when setting this up i was able to put in 3 or 4 command to the terminal, which connect this directory to the github repository, created the .TEX files, created the .venv, created the .vscode/settings.json with some preloaded files to hide.

5 GIT

GIT is a version control system that allows you to track changes in your code and collaborate with others. it is usefull for managing code changes and collaborating with others.

5.1 Git Commands

```
echo "# Example_Project" >> README.md
git init
# Initialize a new Git repository

git add README.md
# Stage the README file for commit

git commit -m "first commit"
# Commit the staged changes with a message

git branch -M main
# Rename the default branch to main

git remote add origin git@github.com:UserName/Example_Project.git
# Add the remote repository

git push -u origin main
# Push the changes to the remote repository

git remote add origin git@github.com:UserName/Example_Project.git
# Add the remote repository

git branch -M main
# Rename the default branch to main
git push -u origin main
# Push the changes to the remote repository
```

In your degree you will use git to manage your code, track changes, and collaborate with your peers on various projects.

6 Fun with the Terminal

Here are some fun things you can do with the terminal: sudo apt install cowsay sudo apt install fortune sudo apt install cmatrix sudo apt install fastfetch or others sudo apt install sl sudo apt install wtf